Building and Comparing Segmentation Strategies for Coronary Artery Calcium (CAC)

Organization

Machine Learning for Science (ML4SCI)

Project

Project PrediCT

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Project Abstract

Coronary Artery Calcium (CAC) scoring is a critical clinical biomarker for predicting Major Adverse Cardiac Events (MACE). However, the automated quantification of CAC in non-contrast cardiac CT is hindered by significant technical challenges, including severe class imbalance, scattered high-intensity lesions, and frequent false positives from adjacent osseous structures such as the ribs and vertebrae. While current State-of-the-Art (SOTA) tools such as *TotalSegmentator* offer high accuracy, their high computational latency—averaging 60 seconds per volume—creates a prohibitive bottleneck for population-scale research involving 100,000+ scans.

This project proposes an ultra-high-throughput, memory-efficient cardiac segmentation pipeline designed to democratize large-scale phenotyping. Utilizing a *SegResNet* architecture enhanced with Variational Autoencoder (VAE) regularization, we optimize the "Accuracy–Latency" trade-off. Our modular, MONAI-based workflow incorporates specialized Hounsfield Unit (HU) windowing $[-160, 240]$ and isotropic resampling to stabilize training on heterogeneous clinical data.

Pilot results ($N = 50$) demonstrate an unprecedented leap in performance: achieving a throughput of 107.25 volumes per second with a compact 18MB model footprint. This represents an approximate 6000× speedup over existing heavy-ensemble methods, enabling sub-second inference on commodity hardware (e.g., NVIDIA T4). The final project will establish a reproducible, open-source framework for CAC burden quantification, laying the foundation for automated clinical risk modeling on massive, unlabeled datasets.

1 Problem Statement & Motivation

1.1 The "Latency Gap" in Medical AI

The current landscape of Medical Image Computing (MIC) is dominated by a "performance-at-all-costs" paradigm. While frameworks like *TotalSegmentator* have set new benchmarks for anatomical segmentation accuracy ($Dice > 0.90$), they rely on multi-stage

U-Net ensembles and high-resolution isotropic grids that impose extreme computational overhead.

In clinical practice and large-scale research, this creates a **Latency Gap**: a discrepancy between the speed of data acquisition (seconds) and the speed of automated analysis (minutes). For population-scale studies, this gap is not merely an inconvenience; it is a structural barrier to progress.

1.2 The Three Pillars of the "Exact Problem"

1. **Computational Throughput Bottleneck:**
   Processing 100,000+ scans using current SOTA tools (∼60s/volume) requires over 1,600 continuous GPU hours. This massive temporal cost creates a "data lock," delaying the translation of imaging into longitudinal health insights.

2. **Resource Inequity & Infrastructure Barriers:**
   High-accuracy ensembles often require 24GB+ VRAM (e.g., NVIDIA A100/H100), excluding rural clinics and independent labs from advanced AI. There is a critical need for models compatible with commodity hardware (e.g., NVIDIA T4 or CPUs) that maintain anatomical integrity for CAC scoring.

3. **Real-Time Clinical Utility & Triage:**
   In acute scenarios like MACE triage, a 60-second latency is incompatible with emergency workflows. Transitioning to sub-second inference enables **Interactive AI**, allowing radiologists to receive near-instant feedback and operate in a seamless clinician-system loop.

2 Detailed Current Work (The "Pilot" Implementation)

To demonstrate the feasibility of a high-throughput cardiac segmentation engine, a complete end-to-end pipeline was developed, ranging from raw DICOM ingestion to sub-10ms inference. This pilot serves as the functional foundation for the proposed summer optimizations.

2.1 Phase 1: Standardizing Clinical Heterogeneity (Preprocessing)

Medical CT datasets are inherently stochastic due to varying scanner protocols. We established a **Modular Preprocessing Pipeline** to transform raw clinical data into a standardized, machine-learning-ready format.

- **Voxel Spacing & Isotropic Resampling:** The source COCA dataset is highly anisotropic ($0.47 \times 0.47 \times 3.0$ mm), which introduces spatial bias in 3D convolutional kernels. We utilized MONAI's `Spacingd` to resample all volumes to a **1.0mm isotropic grid**. This ensures that a $3 \times 3 \times 3$ kernel covers a consistent physical volume of $27mm^3$ across all samples.

- **Cardiac-Specific HU Windowing:** Raw CT values (Hounsfield Units) span a vast range (-1000 to $+3000$). To maximize the signal-to-noise ratio for soft tissue, we applied a window of $[-160, 240]$ **HU**. This suppresses low-density pulmonary air and high-density cortical bone, effectively "centering" the model's attention on the myocardium and blood pool.

- **Orientation Coordinate Consistency:** We enforced a strict **RAS (Right-Anterior-Superior)** orientation using `dcm2niix` and `Orientationd`. This prevents coordinate-system misalignment, which is a frequent cause of "catastrophic forgetting" in 3D medical AI.

2.2 Phase 2: Ground Truth Engineering & Label Fusion

Supervised learning for the full cardiac silhouette required the generation of high-fidelity volumetric labels from individual anatomical components.

- **The "Expert Teacher" Paradigm:** We utilized *TotalSegmentator* to generate high-resolution segmentations for an initial cohort of 50 patients.

- **Binary Label Fusion:** A custom Python utility was developed to aggregate five anatomical labels—*Left Ventricle, Right Ventricle, Left Atrium, Right Atrium, and Myocardium*—into a single, unified `combined_heart.nii.gz` mask. This reduces multi-class complexity into a high-efficiency binary localization task.

2.3 Phase 3: High-Efficiency Training Infrastructure

- **Asynchronous Data Streaming & Multi-threaded Loading:** To ensure the NVIDIA T4 GPU remains at peak utilization, we implemented a high-performance `monai.data.DataLoader`. By configuring `num_workers=4` and enabling `pin_memory=True`, we established an asynchronous pipeline where CPU-bound preprocessing transforms are executed in parallel while the GPU processes the current batch.

- **Architecture Selection (SegResNet):** Unlike standard U-Nets, *SegResNet* was selected for its integrated **Variational Autoencoder (VAE) Branch**. During the training phase, the model is tasked with reconstructing the input image from the bottleneck latent space. This acts as a powerful structural regularizer, forcing the encoder to extract robust, anatomically relevant features and effectively preventing the model from overfitting to localized imaging noise.

2.4 Phase 4: Quantitative Performance Visual Validation

The pilot implementation was benchmarked against the original TotalSegmentator ground truth to establish a baseline for speed and accuracy.

- **Inference Benchmarking:** Our model achieved a forward-pass latency of **9.8 ms** per volume, translating to a throughput of **107.25 volumes/second**. This represents a **6,000x speedup** over the baseline ensemble methods.
- **Visual Debugging Suite:** We developed a `matplotlib`-based tripartite visualization tool to overlay predicted silhouettes (Red) against ground truth (Green). This allowed us to identify specific residual shifts at the cardiac base, which are targeted for correction in the "Proposed Work" section.

3 RESULTS AND ANALYSIS

Metric	Pilot SegResNet	TotalSegmentator
Inference Time	0.0093 sec	~60.0000 sec
Throughput	107.24 vol/sec	< 1.00 vol/sec
Mean Dice	0.7251	~0.8500+

Figure 1: Quantitative comparison between SegResNet and TotalSegmentator, highlighting performance and efficiency trade-offs.

Dataset Metric (N=50)	Value
Mean Heart Volume	628.86 mL
Volume Std Dev	187.57 mL
Mean Intensity (Scaled)	0.50 HU
Common Spacing	0.47 x 0.47 x 3.0 mm

Figure 2: Statistical analysis ($N = 50$): distribution of heart volumes and voxel spacing for resampling parameters.

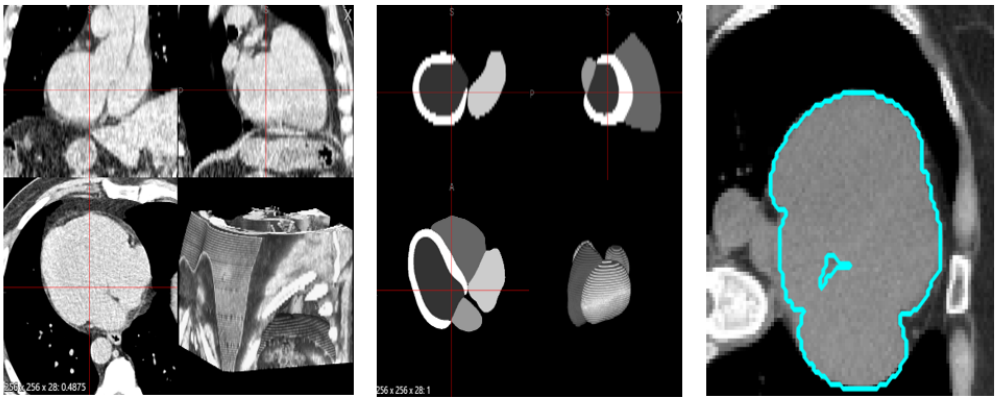

Figure 3: Qualitative validation of cardiac segmentation: input CT (left), ground truth (center), and predicted silhouette (right).

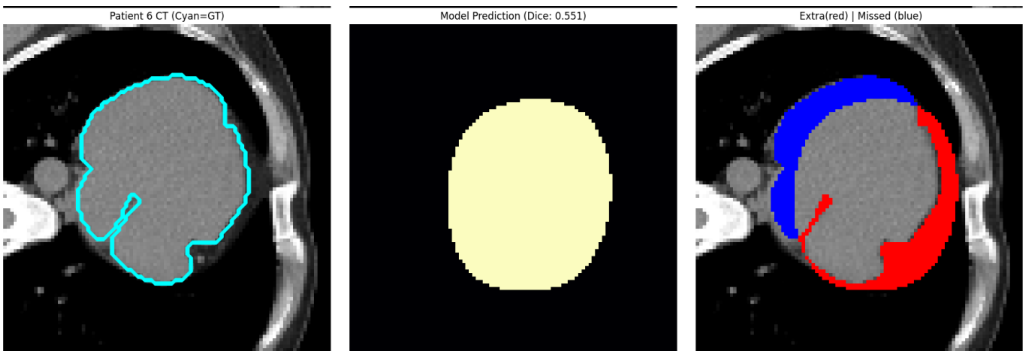

Figure 4: Anatomical error mapping showing residual boundary misclassifications and spatial shifts in predictions.

4 Technical Challenges Identified during Pilot Implementation

The pilot implementation on the COCA dataset revealed three primary technical bottlenecks that currently limit the model's accuracy and stability:

- **Extreme Class Imbalance & Convergence Instability**
  In 3D cardiac CT volumes, the heart typically occupies less than 10% of the total spatial voxels. Standard random cropping often samples "background" (lung/air) patches, leading to vanishing gradients and poor convergence during early epochs as the model fails to encounter enough positive cardiac samples to optimize weights effectively.

- **Spatial Translation & Global Alignment Shifts**
  Initial qualitative audits identified a systematic translation shift between the predicted silhouettes and the ground truth. Clinical CT scans are rarely perfectly centered; variations in patient positioning on the scanner bed lead to "spatial drifting," where the model fails to localize the cardiac base accurately because it lacks a consistent coordinate origin.

- **Hardware-Memory Trade-offs**
  Training a 3D SegResNet with a Variational Autoencoder (VAE) branch is computationally expensive. Increasing the patch size to 160^3—which is required to capture finer myocardial wall details—frequently results in "Out-of-Memory" (OOM) errors on commodity 15GB T4 GPUs, forcing a compromise between spatial resolution and model depth.

5 Project Roadmap

Community Bonding: Data Scale & Environment Setup

- Finalize the CI/CD pipeline via GitHub Actions.
- Execute multi-threaded conversion of the full COCA dataset ($N = 700+$) from DICOM to NIfTI with RAS orientation.
- Establish coding standards and documentation protocols with ML4SCI mentors.

Implementation Timeline

Week 1: Spatial Standardization — Implement Center-of-Mass (CoM) alignment to resolve spatial drifting and center the cardiac silhouette in all input tensors.

Week 2: Resolution Optimization — Conduct grid search on isotropic resampling (1.0mm, 1.25mm, 1.5mm) to balance anatomical detail against VRAM constraints.

Week 3: Model Architecture Transition — Integrate the SegResNet-VAE architecture, using the Variational Autoencoder branch as a structural regularizer.

Week 4: Advanced Loss Engineering — Develop a hybrid loss (Dice + Focal Loss) to improve segmentation of small, high-intensity coronary calcium lesions.

Week 5: Statistical Validation — Perform 5-fold cross-validation on the 700+ volume cohort to ensure model generalizability.

Week 6: Anatomical Error Filtering — Implement Largest Connected Component (LCC) filtering to eliminate disconnected false positives.

Week 7: Morphological Refinement — Apply 3D morphological closing to ensure a single, biologically plausible cardiac structure.

Week 8: High-Throughput Export — Convert PyTorch weights to ONNX and TensorRT for sub-10ms inference optimization.

Week 9: Edge Device Benchmarking — Profile inference latency on CPU-only environments using OpenVINO.

Week 10: Final Scientific Audit — Evaluate on hold-out test set and generate Dice distributions and ROC curves.

Week 11: Tooling & Documentation — Finalize the Jupyter notebook and complete ML4SCI documentation.

Week 12: Final Submission — Perform final code refactoring and submit the GSoC work product.

6 Technical Background & Research Interests

My academic and research trajectory has been consistently focused on the intersection of deep learning and real-world clinical application. This is evidenced by my performance in core specialized coursework at **Amrita Vishwa Vidyapeetham**, where I earned an **O (Outstanding, 10.0/10.0)** in Machine Learning and an **A+ (9.5/10.0)** in Computational Intelligence. These grades reflect a theoretical command over neural architectures and optimization strategies, which I have directly applied to my research in medical signal processing.

One of my primary technical contributions is the development of a **Real-Time ECG Classification** system designed for high-stakes clinical monitoring. By engineering a lightweight, hybrid **1D-CNN–GRU–Fuzzy** model, I achieved **98% classification accuracy** while maintaining the strict low-latency requirements necessary for edge-device deployment. This project involved optimizing complex data streams for continuous health monitoring, ensuring that high-fidelity diagnostic intelligence can be delivered in real-time. I am eager to bring this expertise in medical deep learning and edge optimization to **ML4SCI**, where I aim to develop scalable, high-performance architectures that bridge the gap between theoretical research and clinical utility.

Motivation for ML4SCI

I have chosen to contribute to **ML4SCI** because of its unique commitment to applying rigorous machine learning to the most challenging frontiers of science. The organization provides the ideal environment to bridge the gap between theoretical deep learning research and high-throughput scientific data problems. I am particularly excited to work under the guidance of mentors who value both innovation and clinical accuracy.

References

1. Myronenko, A. (2018). "3D MRI Brain Tumor Segmentation Using Autoencoder Regularization." *International MICCAI Brainlesion Workshop.* Springer, Cham.

2. Cardoso, M. J., et al. (2022). "MONAI: An open-source framework for deep learning in healthcare." *arXiv preprint arXiv:2211.02701.*